Languages in the usnewsmap index: status, October 2026

Pages, text sources, OCR quality and search for the 29 languages Chronicling America catalogs

October 2026

This report grew out of an earlier one on Japanese-language newspapers (merged in #195, in `docs/reports/2026-10-japanese-newspapers`, now renamed to this directory). The Japanese material is now one section among the others.

Summary

The live index, version `pages-v20261006-2` (published 7 October 2026), holds 23,765,561 pages from 4,691 titles. Chronicling America catalogs languages per title, and its titles list 29 languages. Titles that list English hold 22.8 million pages; the other 28 languages range from German's 693,713 pages to Creek's 4. Every language except Japanese is searched in LoC's OCR. LoC has no Japanese text for these papers, so Japanese is searched in our own OCR. A 10% sample of LoC's OCR (2,371,337 pages) finds the most damage in Serbian, Russian and Czech, in German before the 1920s and in English from the 1840s to the 1880s. Part of the high scores for Polish, Czech, Slovak, Serbian, Russian, Finnish, Lithuanian and Yiddish may come from the way the audit counts damage, which a hand check would settle. The five indigenous languages have no word list and aren't scored. The titles cataloged as Hawaiian are English-language papers, and the one Hebrew title's Hebrew-script text looks like Yiddish.

Language	Pages	Titles	Text from	Damage: median; badly damaged	Search	Status
English	22,787,429	4,515	LoC; American Stories being added	0.071; 8.1%	Standard	Worst 1840s to 1880s; five batches almost all badly damaged
German	693,713	108	LoC	0.105; 7.1%	Standard	High damage 1850s to 1910s; Fraktur pilot proposed
Spanish	648,027	88	LoC	0.084; 4.1%	Standard	26% of pages in Spanish titles are English
French	201,287	49	LoC	0.075; 10.3%	Standard	High damage in the 1900s and 1910s
Polish	169,132	21	LoC	0.178; 16.8%	Standard	High in every decade; hand check
Italian	136,111	23	LoC	0.114; 4.1%	Standard	3,893 sampled pages detected as Spanish: *La Gaceta* 3,877, *El Federal* 15,

Language	Pages	Titles	Text from	Damage: median; badly damaged	Search	Status
						Boletin Obrero 1; not manually verified
Yiddish	132,828	6	LoC	0.160; 13.8%	Hebrew script; old and modern spellings are different words	Word list may not fit older spelling
Danish	87,917	4	LoC	0.146; 20.4%	Standard	5,827 sampled pages detected as English: *St. Croix Avis* 5,823, *Danskeren* 3, *St. Paul Tidende* 1; not manually verified
Czech	61,929	10	LoC	0.212; 39.2%	Standard	High, falling over time; hand check
Norwegian	49,414	9	LoC	0.084; 3.0%	Standard	Low damage in every decade
Serbian	48,710	1	LoC	0.257; 52.7%	Cyrillic and Latin spellings are different words	Highest damage; hand check
Japanese	32,213	27	Ours (NDLOCR-Lite); LoC has none	Not scored	Own index, by character	10,305 pages searchable; more with the next release
Finnish	26,353	3	LoC	0.194; 15.6%	Standard	High in every decade; hand check
Hawaiian	24,846	6	LoC	1 page, in a French title	Standard	The titles are English-language papers
Swedish	23,217	8	LoC	0.087; 4.0%	Standard	40% of pages undetermined
Lithuanian	20,554	4	LoC	0.173; 7.8%	Standard	High in every decade
Slovenian	16,236	4	LoC	0.077; 9.6%	Standard	Low damage; one bad title
Hungarian	13,442	3	LoC	0.095; 0.2%	Standard	44% of pages undetermined
Russian	11,930	5	LoC	0.240; 41.2%	Cyrillic	High damage; hand check
Croatian	9,331	3	LoC	0.087; 0.3%	Standard	41% of pages undetermined
Romanian	9,287	2	LoC	0.081; 3.0%	Standard	Many pages undetermined
Choctaw	9,114	2	LoC	No word list	Standard	Sampled pages read as English
Slovak	5,190	1	LoC	0.192; 10.7%	Standard	High, like Czech
Dakota	4,049	2	LoC	No word list	Standard	Dakota text present; unchecked
Navajo	1,295	1	LoC	No word list	Standard	Navajo text present; unchecked

Language	Pages	Titles	Text from	Damage: median; badly damaged	Search	Status
Icelandic	540	1	LoC	0.153; 3.5% (57 pages)	Standard	Small sample
Cherokee	324	2	LoC, including the syllabary	No word list	Syllabary searchable	Unchecked; another collection has more pages
Hebrew	128	1	LoC	0.947 (5 pages)	Hebrew script	The text looks like Yiddish; to check
Creek	4	1	LoC	Not in the sample	Standard	Unchecked

Pages and titles count the titles that list the language, from `/v1/meta`. A title in several languages counts under each, so the column adds up to more than the index. Damage is for LoC's OCR, from the 10% audit, on sampled pages detected in that language: a page of a Spanish title found to be in English counts as English. "Badly damaged" means a damage rate over 0.25. "Standard" search means words are lowercased and accents folded (zeleznice finds železnice), with no stemming, so each inflected form is its own word.

How the numbers are made

The audit. We scored a fixed 10% sample of the pages in index version `pages-v20261003-1`: 2,371,337 of 23.7 million pages, from all 2,989 batches, on 7 October 2026. The 2,426 sampled pages of titles that list Japanese were counted but not scored. For each page the audit does two things.

- **It finds the page's own language.** The candidates are the title's catalog languages plus English. The page's language is the one whose 100 most frequent words make up the largest share of the page's words. A page is *undetermined* when it has too few of those words or two languages come out too close. It is *mixed* when stretches of it fall clearly to two different languages. A page mostly in another script (Hebrew, Cyrillic, Cherokee) takes the title's language in that script.
- **It measures damage in that language.** Of all occurrences of the language's 20 most common words, the damage rate is the share that come out one OCR edit wrong, such as "tbe", "tlie" or "aud" for "the" and "and". A misreading counts only when it isn't itself among the language's 5,000 most frequent words. We call a page *damaged* above 0.1 and *badly damaged* above 0.25. Even good text scores above zero: the lowest median for any language and decade with 100 or more scored pages is 0.043 (English, 1930s).

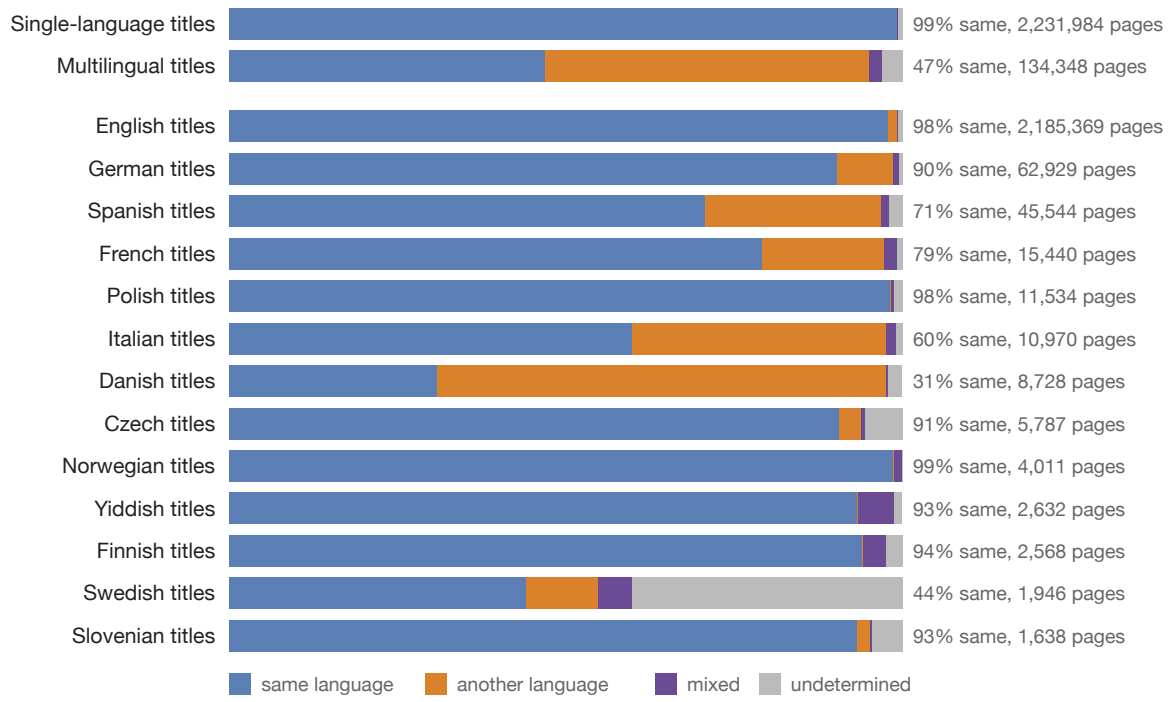
The word lists come from wordfreq, except Yiddish (from Yiddish Wikipedia and Wikisource) and Hawaiian (from the Hawaiian Corpus Project), which wordfreq doesn't cover. Serbian and Croatian share wordfreq's Serbo-Croatian list, and Serbian pages in Cyrillic are scored letter for letter in Latin script. Cherokee, Choctaw, Dakota and Navajo have no list. A 2% sample run the day before, whose pages are all in the 10% sample and which was scored before the Yiddish, Hawaiian and Cyrillic Serbian lists were added, agreed to within a tenth of a percentage point for English and German overall, and within half a point for each decade of English from the 1860s to the 1950s.

Limits of the metric. In heavily inflected languages many real word forms are one edit from a common word and outside the 5,000 most frequent forms, so they count as damage. That would raise the scores for Polish, Czech, Slovak, Serbian, Russian, Finnish and Lithuanian. This explanation is not yet scan-validated. The 15 October #329 lexical screening preserves 20 original-audit pages each for Czech, Serbian, Russian and Croatian controls, and every flagged occurrence. It finds candidate OCR substitutions as well as valid words/abbreviations, but leaves most flags unresolved; it does not establish true-error precision or quantify inflection bias. Slovenian (0.077) and Croatian (0.087) are inflected too and score low, so inflection can't be the whole story. Compare these languages with themselves over time, not with English.

Catalog language and page language. Chronicling America catalogs languages per title, so every page of a title gets the same languages. On the sample:

- **Single-language titles:** 99.3% of pages are in the title's language. Almost all the rest are undetermined (0.7%), and only 0.03% are in another language. For these titles English is the only other candidate (none for English-only titles), so the 99.3% mostly measures how many pages the audit could place.
- **Multilingual titles:** 46.8% of pages are in the first-listed language and 48.3% in a different listed language or English.

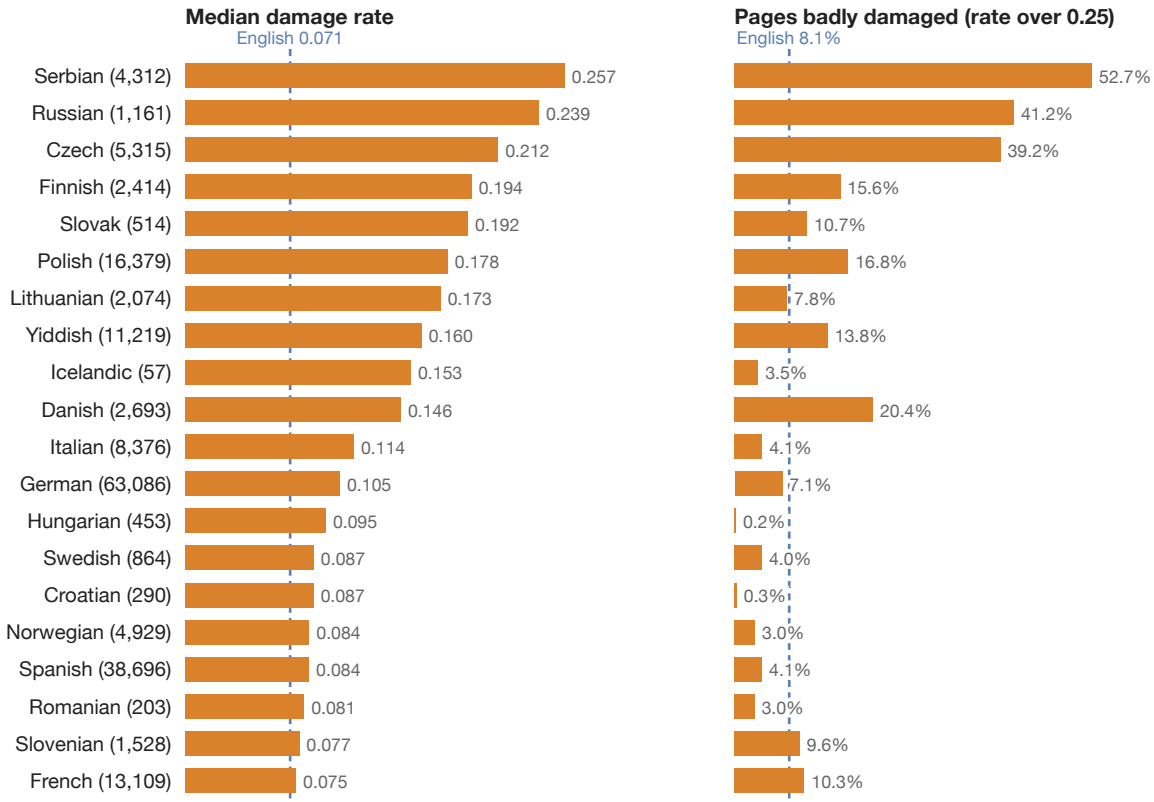
Overall, 3.7% of sampled pages with text are not clearly in their title's first language: 2.8% are in another language, 0.8% are undetermined and 0.16% are mixed. If the sample holds, that is about 880,000 pages, 700,000 of them in another language or mixed. The site's language filter works from the catalog, so `lang=spa` also returns the English pages of Spanish titles. The index has no page-level language yet.



Detected page language against the title's first catalog language, for sampled pages. Below the gap, titles grouped by first catalog language, for languages with at least 1,500 sampled pages.

Search. One analyzer serves every language except Japanese: Unicode word splitting, lowercasing and ASCII folding, with no stemming and no stop words. Accents don't matter (`železnice` and `zeleznice` both find 2,129 pages in Czech titles), but every inflected form is a separate word, and scripts are not folded into each other: in the Serbian title, Србија finds 5,832 pages and `Srbija` 37. Japanese has its own index (see Japanese). The example counts in this report are from `/v1/aggregate` on `pages-v20261006-2` on 7 October 2026.

Languages one by one



Sampled pages by detected language, scored pages in brackets; languages with at least 50. Dashed: English.

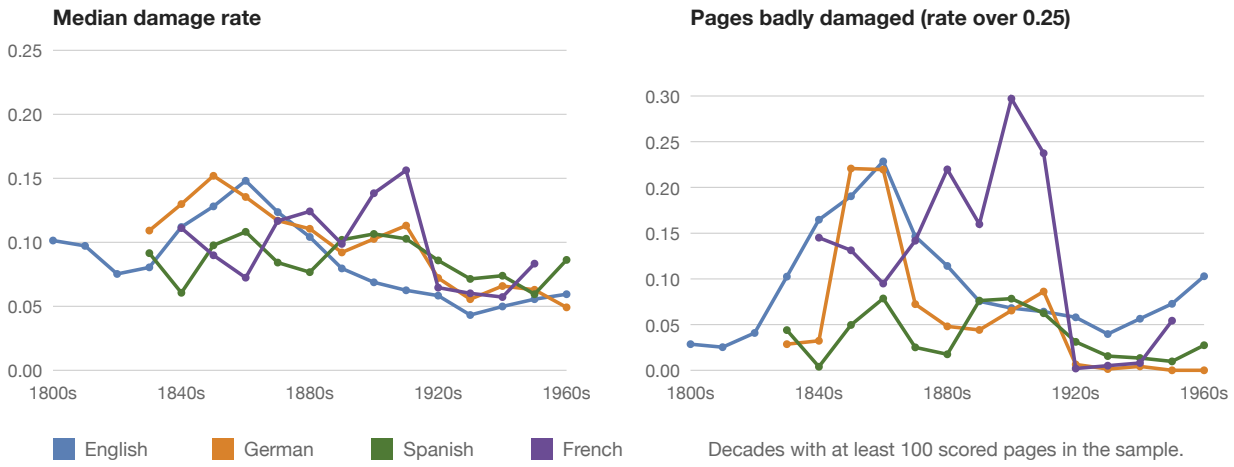
Median damage rate (left) and share of pages badly damaged (right) by detected language, for sampled pages.

Median damage by decade, for decades with at least 100 scored pages in the sample:

Detected language	1840s	1850s	1860s	1870s	1880s	1890s	1900s	1910s	1920s	1930s	1940s	1950s
English	0.11	0.13	0.15	0.12	0.10	0.08	0.07	0.06	0.06	0.04	0.05	0.06
German	0.13	0.15	0.14	0.12	0.11	0.09	0.10	0.11	0.07	0.06	0.07	0.06
Spanish	0.06	0.10	0.11	0.08	0.08	0.10	0.11	0.10	0.09	0.07	0.07	0.06
French	0.11	0.09	0.07	0.12	0.12	0.10	0.14	0.16	0.06	0.06	0.06	0.08
Polish					0.22	0.18	0.18	0.17	0.19	0.17		
Italian						0.15	0.12	0.13	0.07	0.06	0.07	0.06

Detected language	1840s	1850s	1860s	1870s	1880s	1890s	1900s	1910s	1920s	1930s	1940s	1950s
Yiddish								0.18	0.16	0.16	0.17	0.13
Danish				0.11	0.18	0.22	0.15	0.16	0.07			
Czech				0.30		0.32	0.27	0.22	0.19	0.16	0.18	
Norwegian				0.08	0.08	0.09	0.08	0.08	0.08			
Serbian								0.27	0.28	0.30	0.24	0.18
Finnish								0.19	0.22	0.19	0.18	0.19
Swedish							0.09	0.09	0.10			
Lithuanian								0.17	0.19		0.15	
Slovenian								0.10	0.07		0.07	
Hungarian										0.09	0.09	
Russian									0.25	0.23		
Croatian												0.09
Slovak									0.20	0.18		

German, French, Italian and Danish all drop sharply between the 1910s and the 1920s. Polish, Finnish, Yiddish and Serbian don't.



Median damage rate (left) and share of pages badly damaged (right) by decade, for English, German, Spanish and French.

English

The 4,515 titles that list English hold 22,787,429 pages. In the sample, 2,166,233 pages are English: median damage 0.071, 35% damaged and 8.1% badly damaged. Damage peaks in the 1860s (median 0.148; 70% of pages damaged, 23% badly) and falls steadily to 0.043 in the 1930s. In titles that list English first, 97.8% of sampled pages are English and 0.7% undetermined. Most of the rest are pages of titles that list English first and another language as well: 8,763 sampled pages in Yiddish, 6,227 in German, 5,050 in Polish and 4,312 in Serbian.

These are the titles with the highest median damage among all titles with at least 50 scored pages. All but the first are English:

Title	Place	Language	Scored pages	Median damage
Nebraska Staats-Zeitung	Nebraska City and Lincoln, Neb.	German	292	0.62
Corpus Christi Caller and Daily Herald	Corpus Christi, Tex.	English	714	0.55
The Daily Wabash Express	Terre Haute, Ind.	English	191	0.54
Council Bluffs Bugle	Council Bluffs, Iowa	English	140	0.51
The Spirit of the Age	Woodstock, Vt.	English	61	0.51
Weekly Council Bluffs Bugle	Council Bluffs, Iowa	English	58	0.49
True American	New Orleans, La.	English	146	0.48
The Texas Republican	Marshall, Tex.	English	384	0.48
Brownlow's Knoxville Whig, and Rebel Ventilator	Knoxville, Tenn.	English	76	0.47
The Corpus Christi Caller	Corpus Christi, Tex.	English	799	0.46

The five batches with the highest median damage are `nn_kant_ver01`, `txdn_japan_ver01`, `nn_bentham_ver01`, `nn_carson_ver02` and `kyu_frenchie_ver02`. Each has a median damage of 0.46 to 0.52, and 88% to 99.8% of their scored pages are badly damaged. All of their scored pages are English; `txdn_japan_ver01` is LoC's batch name, not a language. All five were among the ten highest in the 2% sample.

American Stories text

American Stories (Dell and others, 2023) re-read Chronicling America scans with a layout model and its own OCR engine, EfficientOCR. Its dataset describes nearly 20 million scans; the current release also includes instances added in March 2025. A precise 2022 snapshot cutoff is not established by the upstream sources. It covers English only ("We do not process foreign language newspapers") and skips ads, tables and regions its classifier marks illegible. We compared it with LoC's text on the audit's sample of English titles for 1865 and 1925 (#205):

	1865	1925
Sampled English pages with an American Stories page-ID match	81%	47%
Median damage, LoC	0.160	0.054
Median damage, American Stories	0.069	0.026
Badly damaged, LoC	26.2%	2.8%

	1865	1925
Badly damaged, American Stories	4.9%	1.4%
Pages added per search term when either text may match (median)	about +15%	about +5%

Its text alone would lose hits, probably mostly in ads, so it goes into the index beside LoC's text, not in place of it: one document per page, and a page counts once whichever text matches. We read by hand 230 matches that only American Stories' text produced: 91% were right, and 98.5% leaving out "radio", where most samples were "Colorado" or "El Dorado" split around "RADIO" (#218). The search side (#220), the ingest side (#222) and a badge saying which OCR a match came from (#226) are merged behind a setting. All 180 years of its text were written to our store by 7 October (about 15 million pages), and a full rebuild with it started that evening; it publishes about 9 October. The 15 October investigation below reproduces the 1925 result exactly: **8,597 of 18,187 pages (47.27%)** have an imported page ID. Most missing pages belong to issues absent from the upstream year archive, especially in later-ingested LoC batches, but some scans are present under colliding page labels. Thus 47% measures the original identifier join, not a verified physical-scan coverage rate.

Why the 1925 join covers only 47% (#327)

Exact historical cohort, not a new index query. The original comparison ran on 7 October 2026, execution `caj-usnm-jaone-prod-puy2mrs`, against `pages-v20261006-2`. On 15 October we reread that version's batch parts and title catalog, selected dates in 1925 and titles whose first catalog language is English, and applied the same fixed hash (`BLAKE2b`, eight-byte digest, integer modulo 10,000 below 1,000). This reproduces all **18,187** sampled pages and **8,597** joined IDs; **9,590** are unmatched. All **91,788** imported 1925 IDs equal the IDs produced by the original placement algorithm over the upstream archive: there is no additional missing-ID loss between that join and our stored import.

The unmatched IDs separate as follows. "Issue" here means the exact LCCN, date and edition, not just the title or day.

Historical sample category	Pages	What the scan-name comparison establishes
Joined page ID	8,597	Same identifier in the original join and imported corpus; 82 are in collision-affected issues and need assignment review
Unmatched; no scan of the issue upstream	8,811	The issue is absent from the 1925 archive, independently of page numbering
Unmatched; issue present, with colliding numbered labels	323	Identifier/join-affected cohort, not 323 proven recoverable pages
Unmatched; issue present, without numbered-label collisions	456	No matching page label; physical absence versus a different labeling error is not exhaustively verified
Total	**18,187**	**8,597 matched; 9,590 unmatched**

Later batches dominate, but "everything after 2022" is not the answer. In LoC's current dataset listing, **7,868** unmatched sample pages (82.0% of the gap), and no matched pages, are in batches with an `ingested` timestamp after 7 June 2023. Another **1,722** unmatched pages are in batches dated on or before then. The date is a useful comparison with the upstream paper's processing period, not a proven freeze date for the current dataset. LoC timestamps can reflect later ingestion/reingestion of a batch version; they do not by themselves establish its first availability. The archive includes matched sample pages from three batches currently dated in February–April 2023 (201 pages), so a strict 2022 boundary is not supported. Across the cohort, **87 batches have zero matches**, accounting for **8,656** missing pages.

The largest missing batch groups are below. Full counts cover **170 batches, 201 titles, all 365 publication dates**, and **9,767 joint batch/title/date groups** in the linked data files; each table reconciles to the same denominator.

Batch	LoC `ingested` date	Sampled	Matched	Missing
`in_enterprise`	2024-04-04	640	0	640
`au_julian`	2024-05-20	601	0	601
`iune_junco`	2025-08-06	541	0	541
`whi_gouda`	2024-09-20	426	0	426
`au_hurston`	2024-02-16	411	0	411
`wyu_laramie`	2022-07-01	380	0	380
`arhi_hauerite`	2025-06-12	349	0	349
`dlc_hine`	2026-01-21	289	0	289
`dlc_erwitt`	2025-12-18	261	0	261
`au_ikenberry`	2024-04-29	225	0	225
`scu_eggplant`	2025-02-20	225	0	225
`ak_goshawk`	2020-12-08	245	25	220

By title, the largest missing groups are *South Bend News-Times* (837 missing, 0 matched), *The Montgomery Advertiser* (636, 0), *The Birmingham Age-Herald* (601, 0), *The Washington Daily News* (550, 0), and *East St. Louis Daily Journal* (541, 0). The gap spans the entire publication year: January has 632 matched / 846 missing pages, and December 834 / 750; it is not a simple newspaper-date cutoff.

The original "duplicates" include different physical scans. The archive has **95,839 distinct (LCCN, date, edition, reel, image) identifiers**, all with parseable names. The current join discards 4,051 entries: **4,050 excess scans at 798 repeated numbered-page labels in 791 issues**, plus one collision involving an inferred `pNone` page. They cannot all be described as duplicate copies of the same image. For example, LoC's 10 January 1925 Indianapolis Times issue has twelve images, `0117`–`0128`. The archive contains all twelve, but eleven are named `p1`. The sampled seq-4 image, `0120`, is present under `1925-01-10_p1_sn82015313_0038334881A_1925011001_0120.json`, yet the name-based join loses it. Its retained seq-1 scan is image `0123`, actually LoC sequence 7. This is a demonstrated join error, not an upstream omission. The **323 unmatched** and **82 matched** sampled pages in collision-affected issues are an affected set, not a corrected coverage estimate. The historical quality/recall figures above have not been rescored

after correcting these assignments. Physical reconciliation and any repair/rebuild are tracked separately in #573; no production data was changed here.

There are also real partial-issue omissions: the 2 January 1925 Alaska Daily Empire issue has eight LoC images (`0007`–`0014`), but the upstream archive contains only its first (`0007`, `p1`). Its sampled seq-3 and seq-7 pages are genuinely absent from the archive. We checked these two physical examples, not every unmatched page in an otherwise present issue. Blindly replacing all page labels with image offsets would misjoin issues whose first images are absent.

Upstream exclusions are not all scan exclusions. The paper excludes foreign-language newspapers and does not transcribe advertisements, tables, and illegible text regions. Those region-level rules explain missing text within a scan, not the 8,811 pages whose entire issue is absent. The English-first cohort includes multilingual titles, so upstream's language policy may also exclude some of these titles; we have not assigned individual omissions to that policy without evidence. The dataset card records a 25 March 2025 update adding previously missing instances; neither its current card nor the paper supplies a batch-by-batch exclusion ledger establishing a 2022 cutoff.

Reproduction and evidence. The retained evidence includes the exact sampled IDs, all imported IDs, the archive's scan names in original order, LoC batch metadata and two physical-image examples, the original audit output, and source hashes in `provenance.json`. Dell and others' scan identifiers are from American Stories (CC BY 4.0). Run this from the repository root, without credentials or network calls, to verify input hashes, recompute the join/classification, and regenerate all grouped tables:

```
python3 docs/reports/2026-10-language-analysis/data/american-stories-1925/analyze.py
```

The batch, title, date, batch/title/date, and ingestion-year tables distinguish the categories above. `classified-sample.csv.gz` retains each page's classification; `identifier-collisions.csv.gz` lists the offending upstream names. The read-only collectors (`collect-sample.py`, `collect-upstream.py`) preserve the collection logic for a fresh check in the existing Azure audit image: the sample collector reads 449 historical curated parts and 434 imported parts, and the upstream collector streams the 1.4 GB archive once. They emit gzip/base64 console chunks and a final decoded-payload SHA-256, not blobs. Decode and verify all numbered chunks before retaining a result; capture the console while the execution replica still exists. All collection ran in Azure under the existing audit job's identity, with no store writes, locks, publication, deployment, or access-policy changes. The retained LoC sources are one dataset listing and two issue JSON responses, requested minutes apart; no per-page request loop was used.

German

The 108 titles that list German hold 693,713 pages. In the sample, 63,086 pages are German: median damage 0.105, 53% damaged and 7.1% badly damaged. In titles that list German first, 90.3% of pages are German, 8.3% English, 0.8% mixed and 0.6% undetermined. Another 6,227 German pages sit in titles that list English first.

Damage stays at 0.09 to 0.15 from the 1840s through the 1910s, with 46% to 78% of pages damaged each decade and 22% badly damaged in the 1850s and 1860s. It drops to 0.072 in the 1920s and 0.056 in the 1930s. The worst German titles are the *Nebraska Staats-Zeitung* (0.62, the highest of any title, with 99% of 292 pages badly damaged), the *Richmonder Anzeiger* (0.43), the *Osage County Volksblatt* and *Hermanner*

Volksblatt in Missouri (both 0.30) and the *Tennessee Staatszeitung* (0.28). The Louisville *Omnibus* has the largest share of undetermined pages of any title: 94% of 111.

Search works as for English: `eisenbahn` finds 165,186 pages in 96 German titles.

Fraktur

Our hypothesis is that much of the damage before the 1920s comes from Fraktur, the blackletter type common in 19th-century German printing, and that LoC's OCR read it poorly. The drop in the 1920s would then reflect the papers' move to roman type around the First World War. We have not checked the typeface of a single sampled page, so both parts are unconfirmed. Danish damage has the same shape (0.22 in the 1890s, 0.066 in the 1920s on 153 pages).

German pages from the 1850s to the 1910s are 57,580 of the sampled pages, so about 580,000 pages in the index. The plan:

1. **Confirm the typeface.** Run Tesseract's Fraktur German model (`deu_latf`) and its roman German model (`deu`) on a sample of about 1,000 pages across the 1850s to the 1910s, fetching LoC's IIIF images as the Japanese job does. Whichever model reads a page better indicates its typeface.
2. **Use the same run as the pilot.** Score the Fraktur output with the damage metric against LoC's text for the same pages. Check about 20 pages by hand for character error rate. Try one or two open Fraktur models trained on historical German (for example models listed by the OCR-D project) against Tesseract's.
3. **Decide with numbers.** Bring the measured gain, the time per page and the cost before deciding on the roughly 580,000 pages.
4. **Ship it like our Japanese OCR and American Stories.** Our text is indexed beside LoC's, one document per page, with a badge saying which OCR a match came from.
5. **Include Danish, Finnish and Lithuanian pages** in the sample, since blackletter type was also used to print those languages (to confirm on our pages).

American Stories won't help here, because its OCR is English-only.

Spanish

The 88 titles that list Spanish hold 648,027 pages. In the sample, 38,696 pages are Spanish: median damage 0.084, 36% damaged and 4.1% badly damaged. Damage is near 0.10 in the 1850s and 1860s and again from the 1890s to the 1910s, when 51% to 55% of pages are damaged, and lowest in the 1950s (0.059).

In titles that list Spanish first, 70.8% of pages are Spanish and 26.2% English (11,919 sampled pages), so a search filtered to Spanish also covers many English pages. Several New Mexico titles have many undetermined pages: *La Revista de Taos* (44% of 351), *La Voz del Pueblo* (33%) and *El Independiente* (30%). `ferrocarril` finds 36,158 pages in 59 Spanish titles.

French

The 49 titles that list French hold 201,287 pages. In the sample, 13,109 pages are French: median damage 0.075, 36% damaged and 10.3% badly damaged. Damage is high in the 1900s and 1910s (0.138 and 0.156, with 30% and 24% badly damaged) and falls to 0.065 in the 1920s, with almost no badly damaged pages. The audit's tables don't split decades by title, so which titles carry the damage of the 1900s and 1910s is unchecked.

In titles that list French first, 79.1% of pages are French and 18.2% English. `chemin de fer` finds 14,299 pages in 37 French titles.

Italian

The 23 titles that list Italian hold 136,111 pages. In the sample, 8,376 pages are Italian: median damage 0.114, 60% damaged and 4.1% badly damaged. Damage is 0.12 to 0.15 from the 1890s to the 1910s and 0.06 to 0.07 from the 1920s on.

In titles that list Italian first, only 59.9% of pages are Italian; 35.5% are Spanish (3,893 sampled pages) and 2.3% English. The 15 October follow-up joined the audit's per-page detections to its snapshot catalog: the 3,893 Spanish detections are in three Tampa papers, *La Gaceta* (`sn86002403`, 3,877), *El Federal* (`sn95047433`, 15), and *Boletin Obrero* (`sn95047434`, 1). The other 13 Italian-first titles in the sample have no Spanish detections. These are detector assignments in the 10% sample, not full-index counts or manually verified page languages (see Mixed-language title attribution). `ferrovia` finds 10,235 pages in 21 Italian titles.

Polish

The 21 titles that list Polish hold 169,132 pages. In the sample, 16,379 pages are Polish: 11,329 in titles that list Polish first and 5,050 in titles that list English first. Median damage is 0.178, with 88% damaged and 16.8% badly damaged. It stays between 0.17 and 0.22 in every decade from the 1880s to the 1930s, with no drop after 1920. In titles that list Polish first, 98.2% of pages are Polish.

For Polish the open question is how much of the score is real damage and how much is inflected word forms counted as damage (see How the numbers are made). `kolej` finds 14,065 pages in 21 Polish titles.

Yiddish

The six titles that list Yiddish, five in New York and one in Providence, hold 132,828 pages. In the sample, 11,219 pages are Yiddish, 8,763 of them in titles that list English first (in the *Yidishes Tageblat*, 2,046 of 3,173 sampled pages are Yiddish). Median damage is 0.160, with 91% damaged and 13.8% badly damaged. It stays between 0.155 and 0.176 from the 1910s through the 1940s, then is 0.131 in the 1950s. In titles that list Yiddish first, 93.3% of pages are Yiddish and 5.4% mixed with English.

The Yiddish word list is mostly modern spelling, and American Yiddish papers before the 1930s often spelled words the older, German-influenced way (דיא for די), so some real words may score as damage. That is unchecked, and the scores don't drop after the 1920s. Search treats the spellings as different words: די finds 119,663 pages in Yiddish titles and 19,201 דיא.

Hebrew

One title lists Hebrew: the *Rhode Island Israelite* (Providence), 128 pages, cataloged as English and Hebrew. The audit found 5 sampled pages mostly in Hebrew script and scored all of them badly damaged (median 0.947). The audit gives a Hebrew-script page the title's Hebrew-script language, here Hebrew. The search snippets we read from this title are Yiddish in the older spelling (דיא, אונד, האט ער ניט), and the other Providence title, *Der Izraelit* (from 1895), is cataloged as Yiddish. If the pages are Yiddish, the 0.947 is Yiddish text scored against a Hebrew word list, not garbled OCR. We read two snippets, not the five sampled pages, so this needs a look.

Danish, Norwegian, Swedish and Icelandic

Danish. Four titles, 87,917 pages, in St. Paul, Minn., St. Paul, Neb. (*Stjernen*), Christiansted in the Virgin Islands, and Neenah, Wis. In titles that list Danish first, only 30.9% of sampled pages are Danish and 66.8% English (5,827 pages). The 15 October follow-up attributes those English detections to *St. Croix Avis* (`sn84037526`, 5,823), *Danskeren* (`sn86086559`, 3), and *St. Paul Tidende* (`sn90059649`, 1); *Stjernen* (`2019270503`) contributes none. These are detector assignments in the 10% sample, not full-index counts or manually verified page languages (see Mixed-language title attribution). The 2,693 Danish pages have median damage 0.146, with 64% damaged and 20.4% badly damaged. Damage rises from 0.108 in the 1870s to 0.221 in the 1890s, stays near 0.155 in the 1900s and 1910s, and is 0.066 in the 1920s, the same shape as German. *Stjernen* has median damage 0.28, with 68% of 274 pages badly damaged. `hvad` finds 18,089 pages in all four titles.

Norwegian. Nine titles, 49,414 pages, in Minnesota, Iowa and South Dakota. 98.6% of sampled pages in titles that list Norwegian first are Norwegian. The 4,929 Norwegian pages have median damage 0.084 (31% damaged, 3.0% badly), between 0.076 and 0.094 in every decade from the 1870s to the 1920s. The one bad title is the *Sisseton Posten* (Effington, S.D.), at 0.27. `jernbane` finds 11,365 pages in the nine titles.

Swedish. Eight titles, 23,217 pages, in Minneapolis, St. Paul and Red Wing, Minn., and Sioux City, Iowa. In titles that list Swedish first, 44.0% of sampled pages are Swedish, 40.1% undetermined, 10.7% English and 5.1% mixed. The undetermined pages are concentrated in one family of titles: *Skaffaren och Minnesota Stats Tidning* (88% of its sampled pages), *Skaffaren* in St. Paul (81%) and in Red Wing (79%), and *Minnesota Stats Tidning* (81% and 50%). *Svenska Monitoren* (Sioux City) has 12%. The cause is unchecked: heavy damage, short pages or text split between Swedish and English can each leave a page undetermined. The 864 pages the audit could place have median damage 0.087 (4.0% badly). `järnväg` finds 239 pages.

Icelandic. One title, *Vínland* (Minneota, Minn.), 540 pages. All 57 sampled pages are Icelandic, with median damage 0.153 and 3.5% badly damaged, a small sample. Accent folding helps here: `Ameríku` finds 108 pages, including pages where the word reads "Ameriku".

Finnish

Three titles, 26,353 pages, in Astoria, Ore., New York Mills, Minn., and Ironwood, Mich. In titles that list Finnish first, 94.0% of sampled pages are Finnish. The 2,414 Finnish pages have median damage 0.194, with 98% damaged and 15.6% badly damaged, and the median stays between 0.18 and 0.22 in every

decade from the 1910s to the 1960s. Finnish marks case with endings, so the inflection effect described above could be large. `rautatie` finds 1,106 pages, and each case form of the word needs its own search.

Czech, Slovak, Slovenian, Croatian, Serbian and Russian

Czech. Ten titles, 61,929 pages, in Chicago, Cleveland, Baltimore, Omaha, St. Paul and Tabor, S.D. In titles that list Czech first, 90.6% of sampled pages are Czech, 5.6% undetermined and 3.3% English. The 5,315 Czech pages have median damage 0.212, with 96% damaged and 39.2% badly damaged. Unlike Polish, the median falls over time, from 0.30 to 0.32 in the 1870s and 1890s to 0.16 in the 1930s. The worst title is *Pokrok Západu* (Omaha), with median 0.36 and 79% of 1,901 scored pages badly damaged; the batch `nbu_goldenalexanders_ver01`, whose scored pages are Czech, ranks 21st of all batches by median damage (0.37). A search snippet from the 1870s shows real damage around a correctly read word (“ziVtaia v«*ti«*kvi … železnice”).

Slovak. One title, the *Youngstownské Slovenské Noviny* (Youngstown, Ohio), 5,190 pages, cataloged English first. Its 514 sampled Slovak pages have median damage 0.192 (0.197 in the 1920s, 0.177 in the 1930s) and 10.7% badly damaged.

Slovenian. Four titles, 16,236 pages, in Chicago, Cleveland and Duluth. In titles that list Slovenian first, 93.3% of pages are Slovenian and 4.5% undetermined. The 1,528 Slovenian pages have median damage 0.077 and 9.6% badly damaged, most of them in the 1910s (25% badly damaged that decade). One title accounts for much of it: *Narodni Vestnik* (Duluth), median 0.37, with 79% of 169 pages badly damaged.

Croatian. Three titles, 9,331 pages. In titles that list Croatian first, 52.2% of sampled pages are Croatian and 41.4% undetermined, almost all in *Zajedničar* (Allegheny, Pa.). The 290 Croatian pages, all but two from the 1940s and 1950s, have median damage 0.087 and almost none badly damaged.

Serbian. One title, the *Amerikanski Srbobran* (Pittsburgh), 48,710 pages, cataloged English first. Its 4,312 sampled Serbian pages have the highest damage of any language: median 0.257, 99% damaged and 52.7% badly damaged, between 0.24 and 0.30 in each decade from the 1910s to the 1940s and 0.18 in the 1950s. Croatian is scored against the same word list and comes out at 0.087, so the list alone doesn’t explain the Serbian score. The Cyrillic OCR itself and the letter-for-letter conversion to Latin are the candidates, unchecked. Search needs the script the page uses (see Search above); the earliest page with the Latin `Srbija` is from 1924.

Russian. Five titles, 11,930 pages, four in Chicago and one in New York. All 1,067 sampled pages of titles that list Russian first are Russian. The 1,161 Russian pages have median damage 0.240, with 99.9% damaged and 41.2% badly damaged (0.254 in the 1920s, with 52% badly damaged, and 0.232 in the 1930s). `Америка` finds 1,702 pages; the Latin `Amerika` finds none.

Reproducible flag screening for #329 (15 October)

A read-only follow-up selected 20 badly scored original-audit pages each for Czech, Serbian and Russian (`damage_rate > 0.25`) and 20 Croatian controls from all detected scored Croatian pages, using fixed-seed SHA-256 ranking. All 80 stored texts reproduce their original rounded scores. The manifest, annotated occurrences, methods and human handoff preserve the 7,631 flags.

These are provisional agent lexical labels, not a completed scan-based hand audit. Likely-error types were screened in concordances; valid-word decisions are occurrence-specific. Most flags remain unresolved, often apparent line-break fragments. No language's true-error precision has been established.

Language	Pages	Flagged occurrences	Likely OCR errors	Valid / false flags	Unresolved	Precision on classified subset only
Czech	20	2,350	479	133	1,738	78.3%
Serbian	20	2,863	1,332	69	1,462	95.1%
Russian	20	1,475	86	81	1,308	51.5%
Croatian control	20	943	64	28	851	69.6%

The last column is `likely / (likely + valid)`, **not** an estimate for all flags or a language ranking. Even assuming those provisional labels are right, treating all unresolved flags first as valid and then as errors gives very broad ranges: Czech 20.4–94.3%, Serbian 46.5–97.6%, Russian 5.8–94.5%, Croatian 6.8–97.0%. Those are conditional bounds, not confidence intervals or scan-verified results.

False flags include Czech `ul.`/`taj.`/`pí.` and address `Ave.`, Serbian address abbreviations `онт.`/`кан.`, Croatian `Karlo`, `se kosi`, dialectal `kak` and archaic `nu`, and Russian `дол.`, `Дж.`, `Жак` and `сего`. Serbian repeatedly has forms such as `јс` near `је`, `сс` near `се` and `којн` near `који`, consistent with character recognition errors rather than inflection alone. Apparent fragments like `manžel ka`, `mo že` and `вре мя` require scan checks to separate lost hyphens, segmentation and valid text; they must not simply be called bad inflections.

The comparison is confounded by title/date/script/layout: 19 Russian pages are from one title, Serbian has only one title, and the Croatian controls are mostly later pages. **All four languages need fluent-reader scan verification, especially Serbian and Russian**, for which the agent's contextual reading confidence is low. Only three representative whole-page derivatives were viewed; Croatian image/ALTO requests timed out. This does not amount to checking all flags against scans.

For **#296 targeting**, preserve a measured Serbian/Czech/Russian pilot and compare scan-based transcription errors before scaling. Do not change language priorities, calibrate replacement gates, or "correct" the published damage rates using these subset ratios. **#329 remains open** for the human check and full adjudication.

Lithuanian, Hungarian and Romanian

Lithuanian. Four titles, 20,554 pages, in Chicago and Cleveland. In titles that list Lithuanian first, 99.3% of sampled pages are Lithuanian. The 2,074 Lithuanian pages (839 of them in titles that list English first) have median damage 0.173, with 95% damaged and 7.8% badly damaged: 0.174 in the 1910s, 0.194 in the 1920s and 0.145 in the 1940s. `geležinkelis` finds 302 pages.

Hungarian. Three titles, 13,442 pages, in Toledo and Youngstown, Ohio. In titles that list Hungarian first, 35.3% of sampled pages are Hungarian, 43.7% undetermined, 10.8% English and 10.1% mixed. Most of the undetermined pages are in the *Amerikai Magyar Hirlap* (Youngstown): 76% of its 722. The 453

Hungarian pages the audit could place, all from the 1920s to the 1950s, have median damage 0.095 and almost none badly damaged. `vasút` finds 114 pages.

Romanian. Two titles, 9,287 pages: *America* (Cleveland, cataloged English and Romanian) and *Românul American* (Detroit). *Românul American*'s sampled pages are 50.1% Romanian, 22.6% English, 18.7% undetermined and 8.5% mixed. *America* has 83% of 568 sampled pages undetermined. The 203 Romanian pages have median damage 0.081 and 3.0% badly damaged.

Japanese

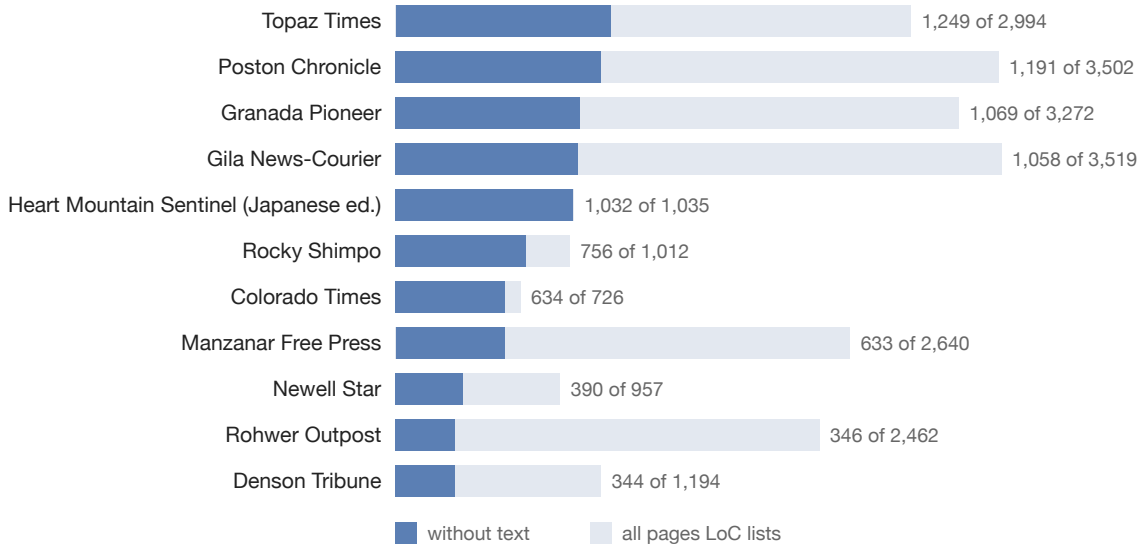
Chronicling America lists 27 titles as Japanese, 32,213 pages including the English pages of bilingual titles. They are mostly the newspapers of the WWII incarceration camps (*Minidoka Irrigator*, *Manzanar Free Press*, the Japanese edition of the *Heart Mountain Sentinel*, *Poston Chronicle*, *Gila News-Courier*, *Topaz Times*, *Granada Pioneer*, *Tulean Dispatch*, *Rohwer Outpost*, *Newell Star* and others) plus two Denver papers, *Rocky Shimpo* and *Colorado Times*. The audit doesn't score them.

The gap at LoC

The page images are fine: LoC serves them at full scan resolution through IIIF (all but 14 pages). The Japanese text is missing.

- In the bulk OCR archives, a Japanese page has an empty ALTO `ocr.xml` and no `ocr.txt`. In one batch, `dlc_ballston_ver01`, 2,433 of 5,791 pages are like that.
- On loc.gov, the page viewer shows "NO TEXT AVAILABLE FOR THIS PAGE", and full-text searches across Chronicling America for 年, 日本 and 真珠湾 return no pages.
- The empty ALTO records what was run. All 124 pages we sampled record `Engine:Abbyy8`: 116 with `Lang:ja` and `Word Count:0`; the other 8 were run as English, which is where the garbled Latin text on some pages comes from.
- Four of the six batches with these papers were ingested in December 2014 and the other two in June 2026. All six were rebuilt in June 2026 and still have no Japanese text. NDNP's technical guidelines have listed Japanese as a supported full-text language since at least the 2016–18 edition.

Comparing LoC's per-title page counts, which include pages without text, with the pages that have text gives about 9,330 Japanese pages across 23 titles with no text. Every other non-English language we checked has its text (German, Spanish, French, Polish, Yiddish, Russian, Serbian, Czech, Hebrew).



Pages with no OCR text at LoC (dark) against all pages LoC lists for the title (light), for the 11 of 23 Japanese titles with the largest gaps. Per-title audit, 4 October 2026 (#135).

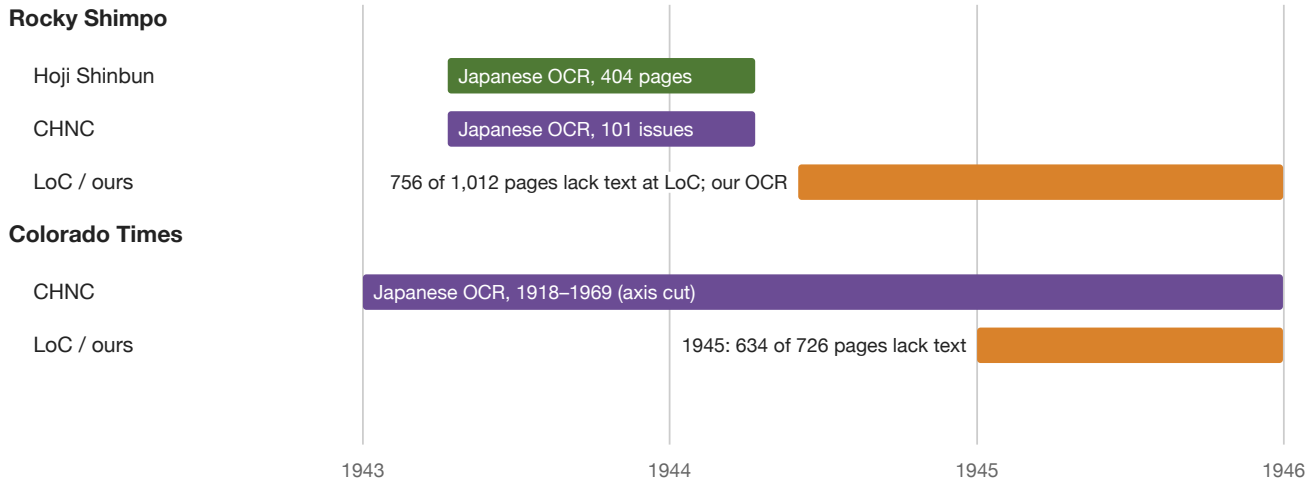
What other collections hold

We looked for Japanese text of the same papers in other online collections on 4, 5 and 7 October 2026. Where a site allowed it, we found the title, searched it for a common word (日本, “Japan”) and for exact lines from LoC’s pages, and compared the dates covered. Several large sites answer automated requests with a bot challenge or disallow them in robots.txt. We did not try to get past either, so for those sites the table rests on their own descriptions and on web search results, and the Japanese text column says “unknown”.

Collection	Holds	Japanese text a search can find	What it means for our OCR
Hoji Shinbun Digital Collection (Hoover Institution)	*Rocky Shimpo* 1943-04-12 to 1944-04-12 (101 issues, 404 pages); camp magazines and directories; not *Colorado Times*, not the camp newspapers	Yes, for what it holds	No overlap: LoC’s *Rocky Shimpo* starts 1944-06-02
Colorado Historic Newspapers (CHNC)	*Rocky Shimpo* 1943-04-12 to 1944-04-12 (101 issues); *Colorado Times* 1918-02-07 to 1969-02-10 (9,973 issues); *Granada Pioneer* and five other Amache titles	Yes for *Rocky Shimpo* (日本: 620 results) and *Colorado Times* (72,008); unknown for *Granada Pioneer* (robots.txt disallows automated access)	No overlap for *Rocky Shimpo*; *Colorado Times* 1945 is already OCR’d there
Densho Digital Repository	Runs of the main camp papers, for example *Poston Chronicle* (670 objects), *Heart Mountain Sentinel* (480), *Gila News-Courier* (436), *Topaz Times* (433),	Unknown (bot challenge). Densho’s published search code indexes catalog fields, not page text	Probably none; to check by hand

Collection	Holds	Japanese text a search can find	What it means for our OCR
	Tulean Dispatch (421), *Manzanar Free Press* (405), *Granada Pioneer* (339), *Minidoka Irrigator* (179)		
Utah Digital Newspapers	*Topaz Times*	Unknown (bot challenge)	To check by hand
Wyoming Newspapers	*Heart Mountain Sentinel*, English edition (issues from 1943 to 1945 seen); no sign of the Japanese edition	Unknown (robots.txt disallows automated access)	To check by hand
ASU Library	*Poston Chronicle* including the Japanese edition; *Gila News-Courier*. OCR'd with Adobe Acrobat, language not stated	Unknown (bot challenge)	To check by hand
Gale, *Japanese-American Relocation Camp Newspapers*	25 camp titles, images from LoC	Unknown (subscription)	Unknown
WOU Repository (Western Oregon University)	238 camp newspaper issues, among them 13 Japanese sections of *Gila News-Courier*, 4 of *Poston Chronicle*, one Japanese *Heart Mountain Sentinel*, and *Topaz Times* 1942-09-17 to 1943-07-24	No: image-only PDFs; 日本 finds 0	None
Internet Archive	*Rohwer Outpost*, 115 items, 1942-10-24 to 1944-01-29; 3 *Manzanar Free Press* items	No: English OCR only; 日本 finds 0 in *Rohwer Outpost*	None
Calisphere (Occidental College collection)	One item each for eight camp titles	Unknown (bot challenge)	Unknown
UA Little Rock Center for Arkansas History and Culture; Arkansas State Archives	*Rohwer Outpost*, *Denson Tribune*, *Communiqué* on paper; *Denson Tribune* on microfilm	No digital copies found	None
National Diet Library Digital Collections	One *Minidoka Irrigator* issue, English	No	None
HathiTrust	No records for these titles	n/a	None

We found no camp newspapers on the California Digital Newspaper Collection, the Arizona Memory Project, the Center for Research Libraries or the Japanese American National Museum's site, but none of the four could be searched automatically. University of Washington Digital Collections has only two maps reprinted from the *Minidoka Irrigator*.



Rocky Shimpo and *Colorado Times*: which collection holds which dates, and whether it has Japanese text. CHNC's *Colorado Times* runs 1918–1969; the axis starts at 1943.

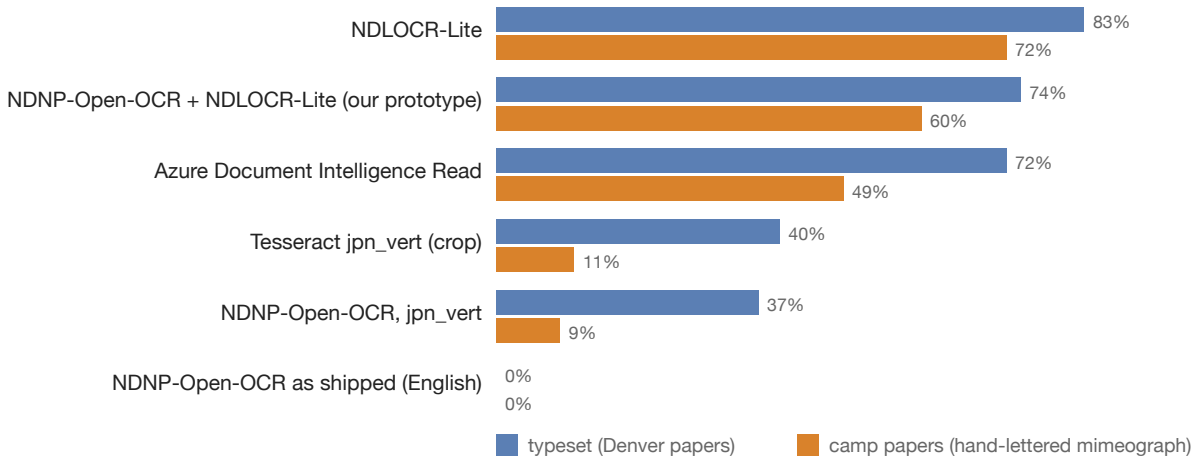
By paper:

- ***Rocky Shimpo*** is split in time. Hoji and CHNC each hold April 12, 1943 to April 12, 1944 (101 issues) with Japanese OCR; LoC holds June 2, 1944 to December 31, 1945 with no Japanese text. Of the collections we checked, our OCR of LoC's 756 pages without text is the only Japanese text for those 19 months.
- ***Colorado Times*** is the one paper where our text duplicates existing work: CHNC has Japanese OCR across 1918–1969. An exact line from the 1945-08-14 issue (皆様お馴染の三光樓) found nothing on CHNC, probably because the two OCR readings differ, which makes the pair useful for comparing quality.
- **The camp newspapers** are not in Hoji, which instead holds camp literary magazines and directories (*Dotō* and *Gyōen* from Tule Lake, an Amache directory) that LoC doesn't have. None of the collections we could search has Japanese text of a camp newspaper. The Internet Archive's *Rohwer Outpost* has English OCR only (no Japanese characters in 108 of its text files), and WOU's issues, including its Japanese sections, are scans with no text layer (none in 44 sampled PDFs). The largest runs, at Densho, Utah Digital Newspapers, Wyoming Newspapers (English edition), CHNC (*Granada Pioneer*) and ASU, could not be searched automatically. Densho's published site code searches catalog fields such as title and description, so a word printed on a newspaper page would not match there even if Densho had OCR'd it. That comes from the code; we did not test it on the live site.

Our OCR

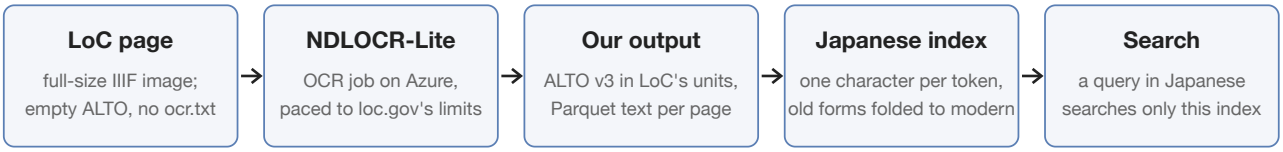
Choosing an engine. We cut 23 reference regions from 124 sampled Japanese pages of 17 titles (1942–45), split between the typeset Denver papers and the camp papers (mostly hand-lettered mimeograph). We scored each engine by bigram F1, the overlap of adjacent character pairs between the engine's text and the reference, which measures whether the words are there, as search needs. NDLOCR-Lite, the National Diet Library's open engine (CC BY 4.0, runs on CPU), scored highest overall, 77% against Azure Document Intelligence's 61%, and beat Azure on 19 of the 23 crops. LoC's reprocessing pipeline, NDNP-Open-OCR (commit `cbe0ca5`), is set up for English only and read every page as English. With Japanese Tesseract models it reached 24%, and with NDLOCR-Lite swapped in as its engine (our local prototype, not proposed

upstream) 67%. The pipeline runs on full pages, and scoring only the text inside each crop costs it some points (#128).



Bigram F1 (overlap of adjacent character pairs with the reference) on the 23 reference crops, typeset and camp papers. NDNP-Open-OCR rows ran on full pages and keep only text inside each crop, which lowers their scores somewhat. The prototype is ours, not LoC's.

The job. For each issue with target pages, the job downloads the full-resolution images from loc.gov, runs NDLOCR-Lite, and writes the text twice: as ALTO in the same units as LoC's empty files, so it could stand in for them, and as one row per page for our index. It paces itself to loc.gov's limits. Targets are the pages LoC ships without text plus pages whose LoC text is empty, short or garbled: 11,058 pages in 3,421 issues.



From a LoC page image to a search result.

Where it stands (7 October 2026).

- **Read:** 11,018 of the 11,058 target pages. Of the other 40, 14 have no image at LoC; the job writes an issue only when every page in it is read, so pages that share an issue with them are held back too.
- **Searchable:** the live version `pages-v20261006-2` has a Japanese index of 10,305 pages (`/v1/meta`). The release read 10,414 OCR'd pages when it started; why 109 fewer are in the index is unchecked. The pages read after that go out with the next release.
- **Search:** a query in Japanese script searches only these pages, one character per token, with old character forms matched to modern ones (戰 finds 戦). 日本 finds 6,269 pages in 21 titles and 真珠湾 443.

Hawaiian

Six titles list Hawaiian, 24,846 pages: the *Polynesian*, *The Pacific Commercial Advertiser* and *Hawaii Holomua* in Honolulu, the *Hilo Tribune*, *The Garden Island* (Lihue) and *The Maui News* (Wailuku). They are English-language papers. Of the 1,415 sampled pages of titles that list Hawaiian first, 98.7% are English and about 1% mix in Hawaiian; none is in Hawaiian alone. Some Hawaiian text is there to search: `ke aupuni` finds 100 pages in five of the titles, the earliest in the *Polynesian* in 1844.

Indigenous languages

Cherokee, Choctaw, Dakota, Navajo and Creek have no word lists, so the audit cannot score them or see their text. Pages of these titles come out as English, undetermined or, for the Cherokee syllabary, mixed. A page shown as English may also carry text in the title's language.

Cherokee

Two titles, the *Cherokee Phoenix* and its successor the *Cherokee Phoenix, and Indians' Advocate* (New Echota, Ga.): 324 pages in 82 issues, 1828-03-06 to 1833-10-19. LoC's OCR has the Cherokee syllabary as Unicode, and search finds it: 𐭂𐭆𐭘 finds 122 pages, 𐭂𐭆𐭘 𐭅𐭅𐭉𐭖0𐭉𐭉 61, 𐭍𐭖 38 and 𐭃𐭋𐭂𐭆𐭘 21. English searches work as well: `removal` finds 109 pages and `new echota` 111. Of 33 sampled pages, 27 show as English and 6 as English mixed with Cherokee.

Georgia Historic Newspapers holds both titles from February 1828 to May 1834, 1,023 pages. Its extra pages, about 700, have English OCR only: all 124 of its pages that match 𐭂𐭆𐭘 fall on LoC's 82 issues. Its page and OCR views sit behind a Cloudflare challenge, and its robots.txt disallows `/data/` and `.txt` and `.xml` files, so we have not compared texts.

New Echota was mapped 84 km from New Echota, near Johns Creek in Fulton County, because the first title's LoC record carries the point of a different New Town. A place override fixes it (#223, #224), and the site moves the point with the next release.

Choctaw, Dakota, Navajo and Creek

- **Choctaw:** 9,114 pages in 2 titles, the *De Queen Bee* (De Queen, Ark.) and *The Indian Journal* (Muskogee, Okla., 4 pages). 883 of 884 sampled pages show as English. Three Choctaw words we tried (`chahta`, `chihowa`, `yakni`) find no pages, so whether these pages carry any Choctaw is unknown.
- **Dakota:** 4,049 pages in 2 titles, *The Oglala Light* (Pine Ridge, S.D.) and *Dakota Tawaxitku Kin, or the Dakota Friend* (St. Paul, Minn.). 87.2% of 398 sampled pages show as English and 12.8% undetermined. Dakota text is in LoC's OCR: `wakantanka` finds 34 pages in both titles, the earliest in 1850.
- **Navajo:** 1,295 pages in 1 title, *Adahooniłigii* (Phoenix, from 1943). 72.6% of 113 sampled pages show as English and 27.4% undetermined. Navajo text is there too: `diné` finds 902 pages and `naabeehó` 728.
- **Creek:** 4 pages of *The Indian Journal*, none in the sample.

For all five the status is unknown. The next step is a look at a few pages of each: how much of the text is in the language, and how well LoC's OCR read it.

Priorities

1. **German Fraktur pilot**, with Danish, Finnish and Lithuanian pages in the sample (see Fraktur). It covers the most pages: about 580,000 German pages from the 1850s to the 1910s.
2. **Hand check of Czech, Serbian, Russian and Polish**, about 20 pages each, counting real OCR errors against the words the audit flags, to separate real damage from the metric's inflection effect. Finnish, Lithuanian, Slovak and Yiddish could be added at little cost. For Serbian, compare with Croatian, which uses the same word list and scores low. The #329 sample and flagged occurrences for Czech/Serbian/Russian/Croatian are now prepared; fluent-reader scan adjudication remains outstanding, and Polish is outside that sample.
3. **The Hebrew pages:** read the five sampled pages and, if they are Yiddish, score them with the Yiddish list.
4. **Page-level language** where it differs from the title's first language: about 3% of pages, in another language or mixed. In single-language titles the audit found almost no pages (0.03%) in another language. Languages without a word list keep their title language.
5. **Undetermined pages** in the Swedish, Hungarian, Croatian and Romanian titles: find out why so many of their pages can't be placed (19% to 44% by language, 83% in *America*).
6. **The worst English batches:** count how many of their pages American Stories covers before planning a re-OCR.
7. **Japanese:** the hand checks of the sites we could not search, a comparison with CHNC's *Colorado Times*, a Japanese reader's review, and the 109 pages missing from the index (see Open questions).
8. **Indigenous languages:** a look at a few pages of each title, Cherokee first.

Caveats

- **The OCR audit is a 10% sample** of `pages-v20261003-1`; page and title counts are from `pages-v20261006-2`. Per-title figures need at least 50 sampled pages: scored pages for damage, pages with text for the undetermined list. The damage rule catches only one-edit misreadings of common words, so it undercounts heavier errors, and even good text scores above zero. The audit only considers a title's catalog languages plus English, so it cannot find pages in a language the catalog leaves out. The Yiddish and Hawaiian lists are our own, from open corpora; how well they suit 19th- and early 20th-century print is unchecked.
- **Damage and page counts group pages differently.** Damage is by the language the audit detects on a page; page and title counts are by the languages a title lists.
- **The Japanese accuracy figures are agreement with a machine reference.** The reference transcriptions were machine-made and one was checked against its scan. The ranking between engines is better established than any one percentage. Camp papers are harder for every engine: NDLOCR-Lite scores 72% on them against 83% on typeset pages.
- **Other collections were checked only by searching their public websites,** and the coverage table reflects what those sites showed on 4, 5 and 7 October 2026. A search that finds nothing doesn't show that a site lacks the text.
- **Example search counts** are page counts on `pages-v20261006-2` on 7 October 2026. They will change with new batches and with American Stories' text.

Mixed-language title attribution

15 October follow-up (#326). The Italian and Danish title counts above come from the existing 7 October audit export, joined by LCCN to the title catalog for the same index version, not from the audit's ranked subsets of titles. In Azure account `stusnmdffapzy`, the sources are:

Container	Blob	Bytes	Last modified (UTC)
curated	`audit/ocr-quality-v2-pages-v20261003-1-10pct-pages.csv.gz`	51,717,530	2026-10-07 05:42:21
reference	`pages-v20261003-1/titles.json`	1,318,915	2026-10-04 00:46:11

The SHA-256 hashes, in the same order, are:

```
ca9890d21db07108f90454e1b16887a18ee4b962d54329ce7cdb409f8e373514
92368e94c4dc921b3fc014eed965a86501b6b71e2a31c2962b666d63f058cb81
```

The blobs were downloaded read-only using `az storage blob download --account-name stusnmdffapzy --auth-mode login`, with the container and blob names above. With the downloads saved as `pages.csv.gz` and `titles.json`, the following reproduces the title attribution without reading newspaper text:

```python
import collections
import csv
import gzip
import json

with open("titles.json") as source:
    titles = {t["lccn"]: t for t in json.load(source)}
counts = collections.Counter()
totals = collections.Counter()
rows = 0
with gzip.open("pages.csv.gz", "rt") as source:
    for page in csv.DictReader(source):
        rows += 1
        first = page["title_language"]
        if first not in ("dan", "ita"):
            continue
        title = titles[page["lccn"]]
        assert title["languages"] == page["title_languages"].split()
        target = "eng" if first == "dan" else "spa"
        if page["decision"] == "detected" and page["language"] == target:
            counts[first, page["lccn"]] += 1
            totals[first] += 1
assert rows == 2371337 - 2426
assert totals == {"dan": 5827, "ita": 3893}
for (first, lccn), count in sorted(counts.items()):
    print(first, lccn, titles[lccn]["name"], count)
```

All 2,368,911 exported rows were parsed, matching the audit's 2,371,337 sampled pages minus the 2,426 Japanese-title pages omitted from this file. All four Danish-first and all 16 Italian-first sampled titles matched the historical catalog, with no catalog-language mismatches or duplicate page IDs within those cohorts. Every decision bucket reconciled to the committed `data/ocr-quality-v2-10pct.jsonl` language-agreement table:

First catalog language	Detected language counts	Mixed (separate)	Undetermined	No text	Total sampled pages
Danish	Danish 2,693; English 5,827	Danish/English 27	181	1	8,729
Italian	Italian 6,571; Spanish 3,893; English 252	Spanish/English 109; Italian/English 37	108	143	11,113

The selections use the **first-listed catalog language**, matching the report's original claims, not every title that lists Danish or Italian anywhere. Mixed, undetermined and no-text decisions do not enter the 5,827 and 3,893 counts. The snapshot catalog lists *St. Croix Avis* as Danish/English, *La Gaceta* as Italian/Spanish/English, and *El Federal* and *Boletin Obrero* as Italian/Spanish.

Review limit. This follow-up checked the catalog join and exported detector records, not the language printed on the pages. Bounded attempts to retrieve representative LoC OCR text timed out, and an additional item request returned HTTP 403; no text or images were successfully reviewed. In particular, the sole English detection in *St. Paul Tidende* (`sn90059649_1916-10-13_ed-1_seq-7`) has only 55 words. A human content/image check remains open. None of these sampled detector counts should be described as manually verified languages or extrapolated to whole-corpus page totals.

Open questions

1. **Japanese text at the sites we could not search,** checked by hand in a browser: search 日本 within *Topaz Times* on Utah Digital Newspapers, *Granada Pioneer* on CHNC, *Heart Mountain Sentinel* on Wyoming Newspapers (and see whether it has the Japanese edition), ASU's Japanese *Poston Chronicle*, and one Densho camp paper.
2. **A quality comparison with CHNC** on the same *Colorado Times* 1945 pages.
3. **A Japanese reader's review** of the reference crops and a sample of our OCR.
4. **The Japanese index count:** the release read 10,414 pages, and the index has 10,305.
5. **Typefaces of the damaged German pages:** how many are Fraktur, and how much a Fraktur model recovers.
6. **Hand-check the mixed-language detections.** The title attribution is resolved above; the detector's page-language assignments have not been manually verified, particularly the 55-word *St. Paul Tidende* page.

Former question 7 (American Stories' 1925 coverage) is answered in the English section: historical join reproduced, upstream omissions and identifier collisions separated, grouped counts retained. The confirmed collision defect and physical reconciliation are separate follow-up #573.

Sources

- Index contents: the usnewsmap.com API on 7 October 2026, all on `pages-v20261006-2`: `/v1/meta` (index version, pages, titles and `languages`), `/v1/places` (places per language), and `/v1/aggregate` and `/v1/hits` with `lang` (example counts and title names). Search analyzer: `infra/quickwit/pages-index.yaml` and design document 05 §5.
- OCR quality audit: `ja-ocr/quality.py` (metric v2) in the usnewsmap.com repository, run on index version `pages-v20261003-1` at 10% on 7 October 2026 (code at commit `480f8e4`) and at 2% on 6 October 2026 (before the Yiddish, Hawaiian and Cyrillic Serbian word lists were added). Its table rows are in `data/ocr-quality-v2-10pct.jsonl` and `data/ocr-quality-v2-2pct.jsonl` next to this report. Word lists: wordfreq; for Yiddish, Yiddish Wikipedia and Wikisource (CC BY-SA 4.0); for Hawaiian, the Hawaiian Corpus Project (CC0); see `ja-ocr/wordlists/README.md`.
- American Stories: Dell and others, American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers (2023), and the dataset (CC BY 4.0); usnewsmap.com #205 (the 1865 and 1925 measurement, 7 October 2026) and #218 (the plan, the precision check and the rebuild estimate), with pull requests #220, #222 and #226.
- Fraktur plan: Tesseract's trained models (tessdata_best: `deu_latf`, `deu`); the OCR-D project.
- Cherokee: Georgia Historic Newspapers, title and issue pages for the *Cherokee Phoenix* and the *Cherokee Phoenix, and Indians' Advocate*, and its robots.txt, 7 October 2026; usnewsmap.com #223 and #224.
- Japanese OCR progress: the usnewsmap.com status API (`https://api.usnewsmap.com/v1/status`, section `ocr_ja`: 11,018 of 11,058 pages read). The 10,414 pages the release read are those the job had written at 6:21 AM ET on 6 October, as recorded in #187. The Japanese search counts are also recorded in #200.
- usnewsmap.com issues #128 (OCR), #135 (the gap and other collections) and the technical note *Japanese pages: OCR of the pages LoC ships without text* (`docs/notes/2026-10-05-japanese-ocr.md`).
- Library of Congress, Japanese-American Internment Camp Newspapers; NDNP Technical Guidelines 2025–27 and 2016–18; NDNP-Open-OCR.
- Hoji Shinbun Digital Collection, Hoover Institution; Colorado Historic Newspapers. CHNC's holdings: Colorado State Library, CHNC: The Largest Digital Collection of Colorado-based Japanese American Newspapers (8 November 2023; *Rocky Shimpo* 101 issues, 1943-04-12 to 1944-04-12; *Colorado Times* 9,973 issues) and A Window to History: News from Amache (the six Amache titles). CHNC's own title page for *Rocky Shimpo* gives the same dates in web search results; its robots.txt disallows automated access, so we did not fetch it. The Hoji and CHNC search counts are from checks by hand on 5 October 2026 (#135).
- Densho Digital Repository collections ddr-densho-145 (*Poston Chronicle*), 97 (*Heart Mountain Sentinel*), 141 (*Gila News-Courier*), 142 (*Topaz Times*), 65 (*Tulean Dispatch*), 125 (*Manzanar Free Press*), 147 (*Granada Pioneer*) and 119 (*Minidoka Irrigator*). The object counts are from web search results of these pages on 7 October 2026, because the site answers automated requests with a Cloudflare challenge. Search fields: densho/ddr-public, `ddrpublic/ui/models.py`, `SEARCH_INCLUDE_FIELDS`, at commit `b19efae` (25 August 2026).
- WOU Repository, Western Oregon University: item set 11 through the Omeka S API (`https://wou.omeka.net/api/items?item_set_id=11`, 238 items); full-text search for 日本 (`https://wou.omeka.net/api/items?item_set_id=11&fulltext_search=日本`) returns 0 items, and

"Topaz" returns items. We ran `pdftotext` on 44 of its PDFs (all 34 tagged Japanese outside *Topaz Times*, and 10 *Topaz Times*); none has a text layer.

- Internet Archive: collection `rohweroutpost`; full-text search for 日本 within it (`https://archive.org/services/search/beta/page_production/?user_query=日本&page_type=collection_details&page_target=rohweroutpost&service_backend=fts`) returns 0, and "Rohwer" returns 113. We read 108 of its 115 OCR text files (7 returned server errors) and found no Japanese characters.
- ASU News, ASU archivist makes Japanese Internment Camp Collection digitally accessible (8 June 2017).
- Gale, Japanese-American Relocation Camp Newspapers: Perspectives on Day-to-Day Life; its content source is given as the Library of Congress by the Hawaii State Public Library System.
- UA Little Rock Center for Arkansas History and Culture, finding aid UALR.MS.0043, *Japanese American Relocation Center newspapers, 1942-1945* (2.0 linear feet), from the CONTENTdm API (`https://arstudies.contentdm.oclc.org/digital/bl/dmwebservices/index.php?q=dmGetItemInfo/findingaids/5917/json`); the Arkansas State Archives' microfilm inventory entry for the *Denson Tribune* (`digitalheritage.arkansas.gov/inventories-microfilm-newspapers/376/`, seen in web search results; the host did not resolve on 7 October 2026).
- Sites checked on 7 October 2026 that answered automated requests with a challenge or disallow them in robots.txt: Utah Digital Newspapers (HTTP 418 challenge), Wyoming Newspapers and the California Digital Newspaper Collection (robots.txt), ASU's PRISM repository and the Arizona Memory Project (challenge), Calisphere and the Online Archive of California (AWS WAF challenge), the Center for Research Libraries (`dds.crl.edu` robots.txt), and Hoji Shinbun (challenge). The Japanese American National Museum's collections site returned HTTP 522. University of Washington Digital Collections: CONTENTdm API title search for "minidoka irrigator", 2 results, both maps.
- NDLOCR-Lite, National Diet Library.